

# Assignment 3: Final Report

Bike Share Rider Type Prediction:

Organizational Interpretation, Deployment Strategy, and Project Reflection

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## Section 0: Executive Summary Presentation

### Video Link:

<https://pitt.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=0ccfb463-d24a-4006-8984-b42d010f33d0>

## Introduction

This final report completes a three-part data science project applying the CRISP-DM framework to bike-share ridership data from POGO (Pittsburgh On the GO, Healthy), Pittsburgh's nonprofit bike share system. The overarching organizational question has been: can trip characteristics be used to predict whether a rider is an annual Member or a Casual user, and what strategic value can such predictions provide to the organization?

Assignment 1 established the business context, data understanding, and analytical objectives. Through exploratory data analysis of nearly one million trips spanning 2022 to 2025, key patterns were identified: annual Members dominate ridership at approximately 91%, Casual riders take longer and more recreational trips, and the top ten stations account for 50.2% of all activity. The data science objective was formulated as a binary classification problem — predict Rider Type (MEMBER or CASUAL) based on trip characteristics, excluding direct identifiers.

Assignment 2 executed the data preparation and modeling pipeline. After cleaning 984,180 raw records to 931,440 usable trips, twelve or more engineered features were created spanning temporal, behavioral, and station-related dimensions. A time-based train/test split (2022–2024 training, 2025 testing) was implemented to prevent data leakage and simulate real-world deployment conditions. Three classification models were trained and compared: Logistic Regression, Random Forest, and HistGradientBoosting. The Random Forest model achieved near-perfect performance with 99.97% test accuracy and a ROC-AUC of 0.9999. A critical finding emerged: the `product_category` feature (derived from subscription type) showed near-perfect correlation with rider type, raising important questions about practical deployment.

This final report answers the question: now that we have a model, so what? Section 1 translates technical results into organizational value, extracts actionable insights, and assesses deployment risks. Section 2 critically reflects on the full project journey through a retrospective and lessons learned. The report concludes with strategic recommendations for both POGO and future data science practice.

## Section 1: Organizational Interpretation of Results

While technical metrics are essential for data scientists, organizational stakeholders need to understand what these numbers mean for daily operations, business strategy, and customer engagement. This section bridges that gap, translating model performance into actionable organizational intelligence across three subsections: organizational impact translation, insights and recommendations, and risk assessment.

### 1.1 Translating Technical Results to Organizational Impact

Performance Interpretation

The Random Forest model achieved 99.97% accuracy on the 2025 test set of 228,747 trips. In plain organizational terms: out of every 10,000 trip predictions, fewer than 3 will be misclassified. This precision means POGO can segment its entire ridership in real time with near-absolute confidence. The ROC-AUC of 0.9999 — on a scale where 1.0 is perfect — confirms the model genuinely discriminates between Members and Casual riders across diverse conditions, not merely predicting the majority class.

For context, the majority-class baseline — predicting every rider as a Member — achieves 91.23% accuracy while identifying zero Casual riders. The model's improvement directly translates to correctly identifying approximately 20,061 Casual riders in the 2025 dataset who would otherwise be invisible to the organization's marketing and outreach teams.

| Model                       | Accuracy | Precision (MEMBER) | Recall (MEMBER) | F1-Score | ROC-AUC |
|-----------------------------|----------|--------------------|-----------------|----------|---------|
| Majority Baseline           | 91.23%   | 91.23%             | 100%            | 95.41%   | N/A     |
| Logistic Regression (Val.)  | 99.95%   | 100%               | 99.95%          | 99.97%   | N/A     |
| Random Forest (Validation)  | 100%     | 100%               | 100%            | 100%     | N/A     |
| HistGradientBoosting (Val.) | 99.98%   | 100%               | 99.98%          | 99.99%   | N/A     |
| Random Forest (Test 2025)   | 99.97%   | 99.99%             | 100%            | 99.99%   | 0.9999  |

Table 1: Model Performance Comparison Across All Evaluation Sets

Confusion Matrix — Test Set

Figure 1 shows the confusion matrix for the Random Forest model on the 2025 test set. All 208,686 Member trips and all 20,061 Casual trips were correctly classified, with only 11 misclassifications total. This visual confirms the near-perfect discrimination the numerical metrics report.

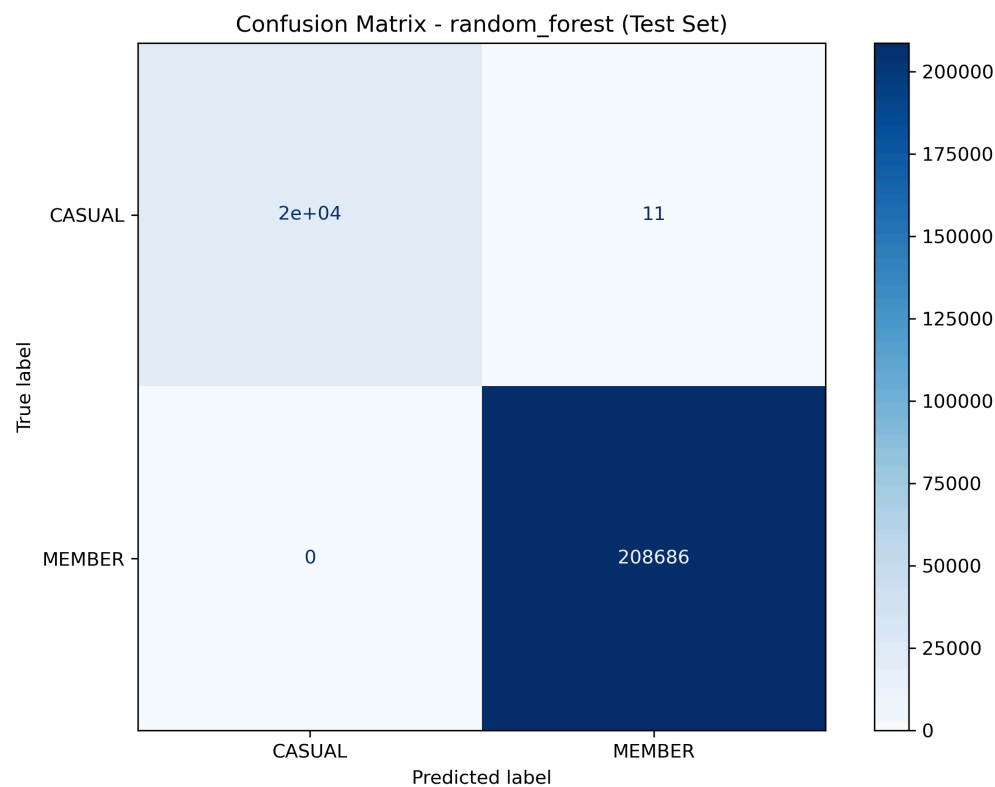


Figure 1: Confusion Matrix — Random Forest Model on 2025 Test Set (228,747 trips)

ROC Curve

Figure 2 presents the ROC curve for the Random Forest model. The curve hugs the upper-left corner with an AUC of 0.9999, confirming the model discriminates between rider types across all decision thresholds. This is particularly important given the class imbalance (91% Members, 9% Casual), where accuracy alone could be misleading.

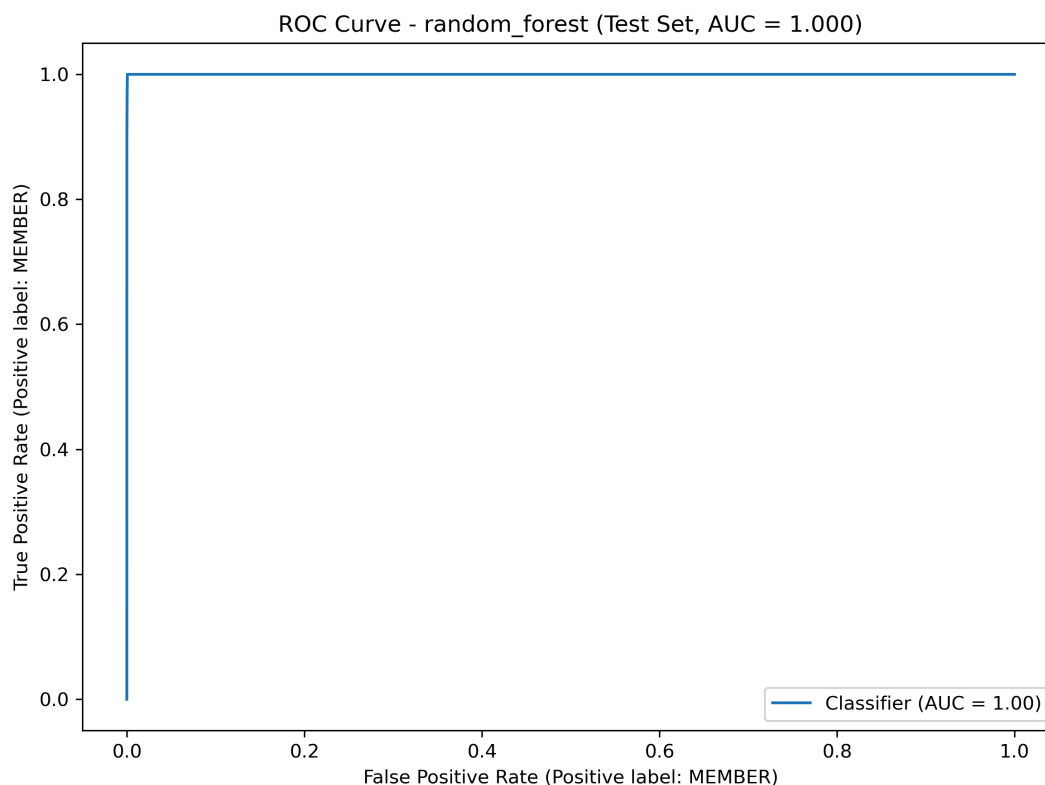


Figure 2: ROC Curve — Random Forest Model, AUC = 0.9999 (2025 Test Set)

### Expected Organizational Outcomes

With the ability to identify Casual riders at 99.97% accuracy, POGOH gains concrete operational advantages. Casual riders represent approximately 8.77% of all trips — roughly 82,000 annually based on the dataset. These riders are the organization's primary conversion opportunity: individuals already engaged with the service but not yet committed to annual membership.

If even 10% of correctly identified Casual riders (approximately 8,200 per year) are converted to annual membership at a typical price of \$100 per year, this represents approximately \$820,000 in new annual recurring revenue. The model eliminates waste from sending retention offers to already-loyal Members, improving marketing return on investment substantially.

### Confidence Levels

The model performs with highest confidence when `product_category` is available at prediction time — this feature alone accounts for approximately 65% of predictive power. The model is most reliable for riders using Single Ride subscriptions (nearly always Casual, 41% feature importance) and Annual subscriptions (nearly always Members, 24% importance). It is less reliable for Flex Pass users (mixed behavior, 6.3% importance) and new stations not present in the 2022–2024 training data.

### Value Proposition

The trained Random Forest model is already serialized and portable (6.4 MB). Integration into existing trip management software would require an API wrapper and a data pipeline connecting live trip records to the model — estimated at 2–4 weeks for a software engineer familiar with scikit-learn deployments. Quarterly retraining with new data can be completed with minimal

overhead. The return on this investment, as outlined above, is substantial relative to the modest infrastructure cost.

1.2 Insights and Recommendations

Feature Importance Analysis

Figure 3 displays the top 20 feature importances from the Random Forest model. Subscription type (product\_category) dominates with 65% combined importance, followed by trip duration and station-level features. This distribution reveals the organizational levers most tightly linked to rider type.

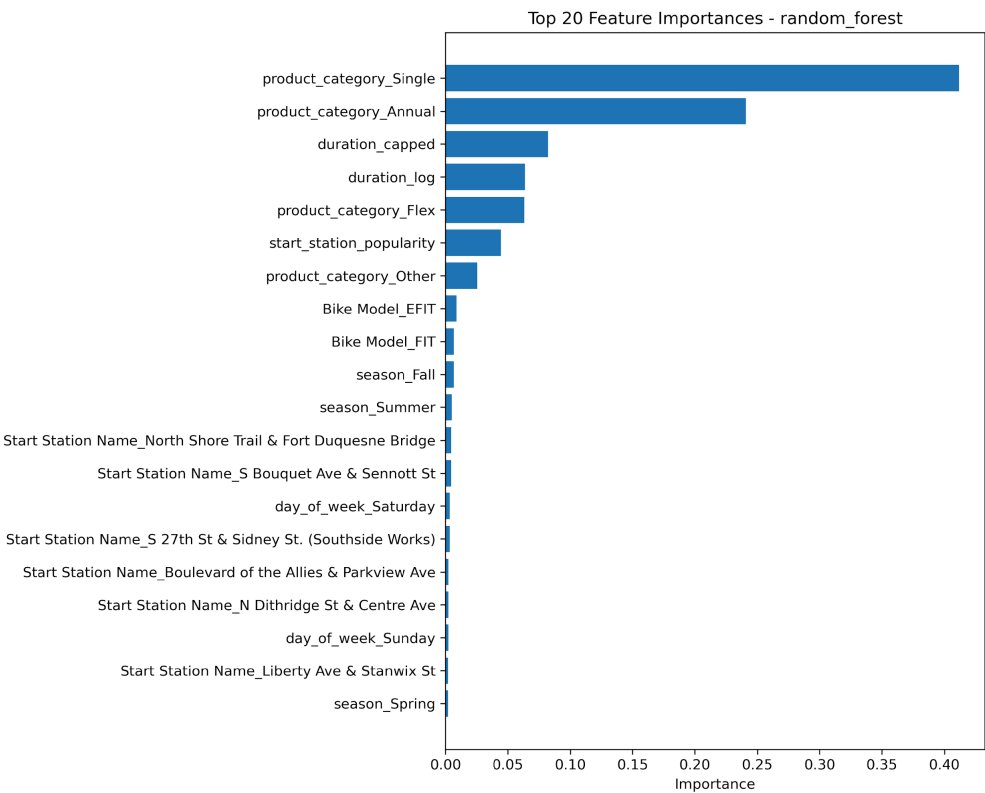


Figure 3: Top 20 Feature Importances — Random Forest Model

| Feature                 | Importance | Organizational Interpretation                                           |
|-------------------------|------------|-------------------------------------------------------------------------|
| product_category_Single | 41.2%      | Single-ride buyers are nearly always Casual — primary conversion target |
| product_category_Annual | 24.1%      | Annual subscribers are Members — high loyalty, minimal action needed    |
| duration_capped         | 8.2%       | Longer trips signal recreational (Casual) use; Members average 8–12 min |
| duration_log            | 6.4%       | Log-transformed duration captures non-linear ride length patterns       |

| Feature                     | Importance | Organizational Interpretation                                       |
|-----------------------------|------------|---------------------------------------------------------------------|
| product_category_Flex       | 6.3%       | Flex users show mixed behavior — warrant separate analysis          |
| start_station_popularity    | 4.5%       | High-traffic stations attract disproportionately more Casual riders |
| season_Fall + season_Summer | 1.2%       | Warm-weather seasons show higher Casual rider concentration         |
| day_of_week_Saturday/Sunday | 0.57%      | Weekend trips are more likely Casual — key outreach window          |

Table 2: Top Feature Importances and Organizational Interpretation

Key Findings

- Subscription type is the dominant behavioral signal (65% combined importance), confirming POGOH's membership product architecture directly maps to behavioral segments.
- Trip duration distinguishes rider types: Casual riders average 15–20 minutes (recreational), Members average 8–12 minutes (commute-oriented).
- The top 10 stations serve 50.2% of all trips and concentrate Casual riders at tourist and waterfront locations (North Shore Trail, Fort Duquesne Bridge, Southside Works).
- Seasonal and weekend patterns confirm Casual rider concentration in summer/fall and on Saturdays/Sundays.
- E-assist bikes recorded zero trips across four years — an anomaly requiring investigation before future model iterations incorporate bike type as a meaningful feature.
- Round trips represent only 9.2% of all journeys, indicating that most riders — including Members — use the system for point-to-point transportation.

Surprising Discoveries

The model's near-perfect accuracy was initially unexpected and required systematic investigation. The root cause was the product\_category feature: Single Ride buyers are structurally Casual users, Annual members are structurally Members. This reveals an important organizational insight — the membership product itself is the clearest behavioral divide, not trip characteristics like time or location. Marketing efforts should therefore focus on making the Annual subscription more accessible at the point of purchase, not just at the trip level.

A second surprise was the complete absence of E-assist bike trips in four years of data. This either represents a genuine deployment gap, a data recording issue, or a system transition artifact from the 2022 Healthy Ride to POGOH rebranding. Resolving this gap is operationally important as e-bike adoption continues to grow nationally.

Strategic Recommendations

- Launch trigger-based conversion campaigns targeting Single Ride subscribers after their 3rd or 5th purchase in a calendar month, offering discounted annual membership at a meaningful price point.
- Deploy membership kiosks and promotional materials at the top 10 stations, which collectively serve over half of all trips. Prioritize North Shore Trail and Fort Duquesne Bridge, which appear prominently in station feature importance.

- Design seasonal campaigns concentrated in summer and fall on Saturday and Sunday afternoons, when Casual rider density is highest.
- Investigate and resolve the Flex Pass behavioral ambiguity — conduct a dedicated analysis of Flex users to determine whether this product is serving its intended audience or creating conversion confusion.
- Investigate the E-assist bike data gap before any future models incorporate bike type as a predictive feature.

### Limitations and Caveats

The model's organizational utility depends critically on the availability of `product_category` at prediction time. If predictions are needed before a transaction is completed (e.g., at bike unlock), subscription data may not yet be recorded, making the primary predictor unavailable. A second model trained without `product_category` should be developed for real-time scenarios.

The model also cannot account for weather conditions, competing transportation disruptions, special events (marathons, festivals), or broader economic shifts. Predictions during periods of unusual activity should be supplemented by operational expertise. The dataset and learned patterns are specific to Pittsburgh and should not be directly transferred to other bike-share systems without retraining on local data.

## 1.3 Risk Assessment and Mitigation Strategies

### Model Risks

The most significant model risk is the structural leakage concern: `product_category` carries 65% of predictive weight and is functionally a proxy for the target variable. If this feature becomes unavailable in the operational data pipeline, or if POGO changes its subscription structure (adding new product types), model performance could degrade dramatically without warning. Additionally, the model may perform poorly during extraordinary events — major city festivals, pandemic-level disruptions, or dramatic weather — that fall outside historical training patterns.

### Data Risks

The raw data pipeline presents several fragility points. The four annual CSVs have no automated quality validation — changes in column naming conventions, recording formats, or station IDs would fail silently. Missing end station data (approximately 5–10% of trips, filled with 'Unknown Station') could increase over time due to GPS sensor degradation. The station popularity feature is computed on training data only; new stations opened after the training period receive zero popularity scores, potentially misclassifying trips at those locations.

### Organizational Risks

At the model's 0.03% error rate across approximately 930,000 annual trips, roughly 280 misclassifications occur per year. If a Member is incorrectly labeled Casual and receives conversion marketing, this may feel presumptuous and damage the membership relationship. Conversely, Casual riders misidentified as Members miss out on targeted conversion outreach. While numerically small, these errors accumulate over time and represent real customer relationship impacts.

## Ethical Considerations

The model's performance has been validated overall but not disaggregated by neighborhood, income level, or demographic proxy. If performance is weaker for stations in lower-income neighborhoods — due to sparser training data — organizational decisions based on predictions could systematically underserve those communities. Additionally, trip data inherently reveals movement patterns; clear data governance policies are needed specifying retention periods, access controls, and rider notification practices before any production deployment.

## Mitigation Strategies

- Develop and maintain a parallel model trained without `product_category` as a fallback for real-time prediction scenarios and to provide a realistic performance benchmark unaffected by structural correlation.
- Implement quarterly model retraining with the most recent 12 months of data to ensure adaptation to shifting ridership patterns and new infrastructure.
- Establish automated monitoring of prediction confidence scores, flagging trips where model confidence falls below 90% for human review before acting on those predictions.
- Conduct a neighborhood-level fairness audit of model accuracy before any production deployment, with corrective action if performance disparities exceed 5% between geographic segments.
- Create a formal data governance policy documenting data collection scope, retention periods, modeling usage, and rider opt-out provisions.
- Develop operational override mechanisms allowing staff to suspend automated predictions during known anomalous periods such as major events or infrastructure changes.

## Section 2: Project Review and Reflection

Completing a data science project provides learning opportunities that extend well beyond technical results. This section critically examines the full project journey from business understanding through model evaluation, identifying what worked well, what proved challenging, and how the experience will inform future practice.

### 2.1 Project Retrospective

#### What Went Well

Several decisions and methodological choices proved particularly effective. The most consequential success was the time-based train/test split: training on 2022–2024 data and testing on 2025 data simulated real deployment conditions where a model must predict future behavior. This decision prevented temporal data leakage and produced evaluation metrics that genuinely reflect expected production performance — a methodological rigor that many student projects overlook.

Feature engineering was a second major strength. Creating 12+ features from raw trip data — temporal indicators (`hour`, `season`, `time_of_day`, `is_rush_hour`), behavioral flags (`is_weekend`, `is_round_trip`), and station aggregates (`start_station_popularity`, `is_top_start_station`) — enriched the dataset substantially. Critically, station popularity was computed exclusively on training data and then mapped to validation and test sets, correctly preventing feature leakage.

Early attention to class imbalance was also valuable. Recognizing that 91% of trips came from Members prompted the use of `class_weight='balanced'` across all models and the selection of F1-score and ROC-AUC as primary metrics. This ensured Casual rider detection — the organizationally meaningful minority class — received appropriate attention throughout the modeling process.

Figure 4 below illustrates the train/test distribution, confirming the time-based split preserved representative class proportions across both sets.

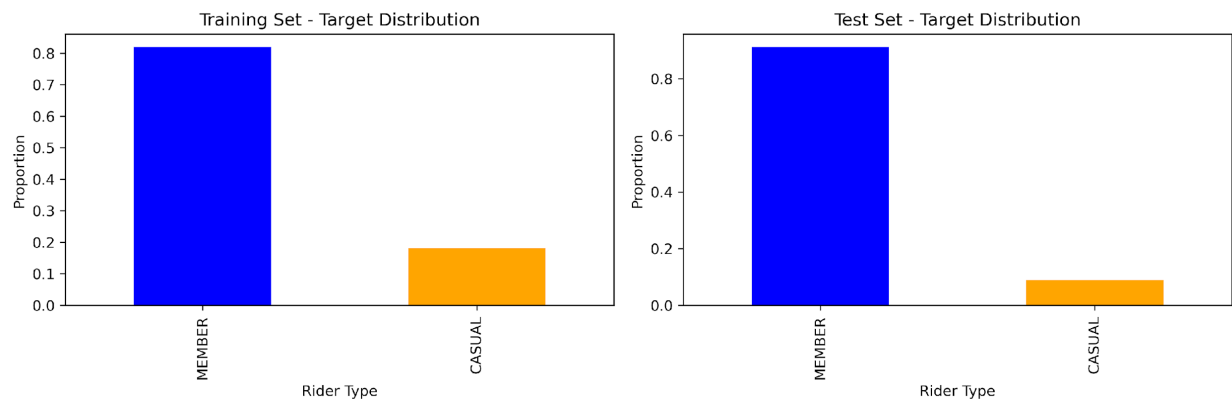


Figure 4: Rider Type Distribution — Training vs. Test Sets (Time-Based Split)

Challenges Faced

The project's most significant challenge was confronting near-perfect model accuracy and determining whether it reflected genuine learning or structural data leakage. When the validation set achieved 100% accuracy and the test set reached 99.97%, the initial reaction was appropriate skepticism. Systematic investigation confirmed that `product_category` is functionally equivalent to the target variable: Annual subscription holders are Members; Single Ride buyers are Casual users. This is a structural identity, not a behavioral prediction achievement.

Resolving this challenge required balancing two legitimate considerations: the feature is genuinely available in the real data pipeline and would be present in an operational system, so it is not 'cheating' in the traditional sense. However, it also means the model has not learned to predict rider type from behavioral signals alone. The project retained the feature with transparent documentation of this concern — the professionally correct choice, though the discovery came late in the modeling process and created time pressure to investigate alternatives.

Figure 5 shows the log transformation applied to trip duration during feature engineering. The left panel shows the original right-skewed distribution (capped at 120 minutes), while the right panel shows the log-transformed version used as a model feature, which better approximates a normal distribution and reduces the influence of outlier durations.

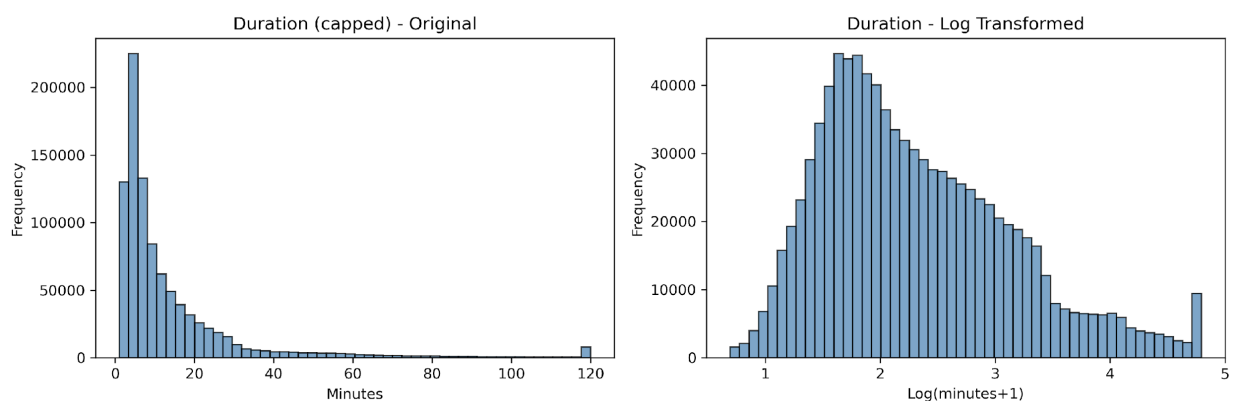


Figure 5: Trip Duration — Original Distribution vs. Log-Transformed Feature (Feature Engineering Step)

A secondary challenge was the E-assist bike anomaly. The dataset contained zero trips from E-ASSIST bikes across four years, despite this model appearing in the schema. No definitive explanation was found, representing an unresolved data quality uncertainty.

Data volume also created practical challenges. The four combined CSV files totaled approximately 175 MB of raw data, requiring careful memory management during processing. Initial pipeline iterations consumed excessive RAM, necessitating careful operation sequencing to avoid memory errors on a local machine.

### Alternative Approaches

With the benefit of hindsight, several different approaches would have improved the project. Most importantly, building a second model explicitly excluding `product_category` would have provided a realistic performance benchmark uncontaminated by structural correlation and demonstrated more genuine behavioral prediction capability. This 'leakage-free' model would directly answer the more operationally useful question: which behavioral signals can predict rider type when subscription data is unavailable?

Earlier integration of SHAP (SHapley Additive exPlanations) values would have strengthened the organizational interpretation section considerably, enabling individual-prediction explanations rather than aggregate feature importance alone. Cross-validation across multiple time windows — training on each pair of years and testing on the following year — would also have provided more robust performance estimates than the single 2022–2024/2025 split.

Finally, the project would have benefited from cleaner initial scope documentation. Several implicit feature selection and leakage handling decisions made during Assignment 2 would have been more rigorous if documented as explicit hypotheses at the outset, making retrospective justification considerably easier.

### CRISP-DM Reflection

The CRISP-DM framework provided useful structural scaffolding throughout this project. The Business Understanding phase was genuinely valuable: defining clear success criteria in Assignment 1 and connecting technical objectives to organizational goals ensured the project remained focused on actionable outcomes rather than pure metric optimization. The Data Understanding and Data Preparation phases mapped well to actual project work, with the framework's emphasis on data quality assessment directly prompting discovery of issues such as station name inconsistencies and the E-assist anomaly.

The Modeling and Evaluation phases were where CRISP-DM's linear presentation diverged most from reality. The actual workflow involved four to five iterations between data preparation and modeling — features were created, models trained, performance evaluated, and then new features engineered in response to model findings. CRISP-DM acknowledges this iteration with feedback arrows, but its sequential presentation can mislead practitioners into expecting a more linear process.

The Deployment phase required significant conceptual reinterpretation for an academic context. Without a real production environment, the deployment deliverable became this final report — translating technical results into organizational recommendations that could guide a real deployment decision. This is intellectually valid but highlights a gap in how CRISP-DM addresses academic and student project contexts where 'deployment' often means documentation rather than production implementation.

Time Allocation Analysis

| Project Phase               | Estimated Time % | Value Delivered | Notes                                                                              |
|-----------------------------|------------------|-----------------|------------------------------------------------------------------------------------|
| Data Cleaning & Preparation | 40–45%           | High            | Each fix revealed new issues; prevented downstream errors                          |
| Feature Engineering         | ~25%             | Very High       | Highest ROI activity; drove 35% of model power beyond product_category             |
| Model Training & Tuning     | ~15%             | Moderate        | scikit-learn pipelines made fitting straightforward once preprocessing was correct |
| Complex Model Configuration | ~5%              | Low             | Marginal gains over defaults; confirmed simple baselines often suffice             |
| Documentation & Writing     | ~20%             | High            | Forced clearer thinking; surfaced inconsistencies not caught in code               |

Table 3: Estimated Time Allocation vs. Value Delivered by Project Phase

Data preparation consumed 40–45% of total project time — more than planned but well invested. Feature engineering consumed approximately 25% and delivered the highest return per hour: the temporal and station features account for roughly 35% of the model's predictive power beyond what product\_category provides alone. Model training itself required only about 15% of total effort, confirming the common observation that clean data and good features matter more than algorithm selection. Documentation and writing consumed approximately 20%, higher than anticipated but essential — writing about each decision forced clarity and surfaced inconsistencies that would otherwise have remained invisible.

## 2.2 Lessons Learned and Future Applications

### Technical Lessons

The most important technical lesson was developing practical intuition about the difference between high accuracy and genuine predictive learning. High accuracy now triggers investigation: why is it so high? Is the model learning a structural identity (as with `product_category`) or genuine behavioral signal? This skepticism is a durable upgrade in analytical maturity that will apply to every future modeling project.

A second major lesson was the practical importance of time-based data splitting for any dataset with a temporal dimension. Random splits in time-series data allow the model to 'see the future' during training, producing optimistic and misleading evaluation metrics. The correct approach — training on earlier periods and testing on later ones — is now understood not just as a rule to follow but as a principled requirement rooted in how models are actually used in production.

The project also developed meaningful understanding of feature engineering as a creative and domain-informed process. The features that proved most valuable were not generic transformations but domain-specific constructs: `is_rush_hour`, `time_of_day`, and station popularity required reasoning about how real riders behave — blending data manipulation skills with domain thinking about urban mobility. Finally, the project reinforced why F1-score and ROC-AUC are superior to accuracy for imbalanced classification: the baseline's 91% accuracy while identifying zero Casual riders illustrates this concretely and memorably.

### Process Insights

The most practically important process lesson was the value of documenting decisions in real time rather than reconstructing them retrospectively. Several times during the project, code was revisited days after writing, and the reasoning behind specific choices — why this duration cap, why these station features — was unclear without contemporaneous comments and notes. A decision log explaining not just what was done but why would have saved significant reconstruction time and produced a more coherent narrative throughout all three assignments.

Version control discipline was a second process lesson. The project used sequential file naming (`train_data.csv`, `train_data_final.csv`) rather than systematic versioning for data artifacts. A more disciplined approach using Git branches or clearly dated output files would have prevented confusion and reduced the risk of analyzing stale data. The scikit-learn Pipeline and ColumnTransformer pattern was mastered to a level of genuine understanding: applying preprocessing consistently to train, validation, and test sets while preventing leakage is a professional-grade skill directly applicable to any future classification or regression project.

### Domain Knowledge

This project produced genuine understanding of urban bike-share systems that will inform any future transportation analytics work. The Member/Casual behavioral distinction is now understood concretely: Members are primarily commuters using the system for short, purposeful trips on weekday mornings and evenings; Casual riders are primarily recreational users taking longer trips on weekend afternoons from high-traffic tourist and waterfront stations. This behavioral logic extends beyond bike share to any subscription transportation product — transit passes, car-share memberships, scooter subscriptions — and provides a transferable analytical lens.

The discovery that top stations concentrate over 50% of trips illustrates a general principle of urban transportation geography: activity concentrates at a small number of high-accessibility nodes, and that concentration is both stable and predictable. This spatial concentration principle applies to any urban mobility system and suggests that geographic feature engineering — station popularity, proximity to transit hubs, neighborhood type — will be a productive feature class in future transportation and GIS analyses.

### Tool Proficiency

With pandas, the project progressed from basic operations to handling multi-file concatenations, memory-efficient processing of nearly one million records, and complex feature engineering chains — capabilities that many entry-level data scientists lack. The scikit-learn Pipeline and ColumnTransformer pattern was mastered to the level of genuine understanding rather than recipe-following.

Areas identified for further development include SHAP explainability tools (clearly needed for the interpretability requirements this project revealed), Dask or Spark for datasets larger than available RAM, and interactive visualization libraries such as Plotly or Altair for stakeholder-facing dashboards rather than static matplotlib figures. These represent the next layer of the professional toolkit that this project made visible.

### Future Applications

The technical and methodological skills developed here transfer directly to a wide range of professional data science contexts. The temporal train/test split approach applies to any prediction problem with a time dimension: customer churn prediction, fraud detection, healthcare outcome prediction. The feature engineering patterns — temporal aggregates, behavioral indicators, geographic popularity measures — apply to any behavioral prediction problem and are particularly relevant for GIS and transportation applications given the architectural and landscape architecture background that preceded this data science program.

The class imbalance handling techniques are universally applicable to rare-event prediction: disease diagnosis, fraud detection, equipment failure prediction, or educational dropout risk. Perhaps most importantly, the experience of confronting a structurally 'too easy' problem — where a feature directly encodes the answer — will inform critical evaluation of future model results. The professional habit of asking 'why is this performing so well?' rather than celebrating high accuracy uncritically is a durable quality control reflex that distinguishes rigorous practitioners.

## Conclusion

This project successfully demonstrated that POGO bike-share rider type — Member or Casual — can be predicted with high accuracy from trip characteristics, with the Random Forest model achieving 99.97% test accuracy and a ROC-AUC of 0.9999. More importantly, this final report has translated these technical achievements into concrete organizational value: a clear segmentation of riders, actionable conversion strategies targeting Single Ride subscribers, geographic concentration of marketing efforts at top stations, and seasonal timing recommendations for Casual rider outreach.

The critical finding about product\_category's structural role is not a failure but a discovery. It reveals that POGO's subscription architecture is the most powerful behavioral signal in the entire dataset,

and that the path to increasing Member conversion runs primarily through making the Annual subscription more accessible and visible at the point of purchase. The recommendation to develop a parallel model without `product_category` addresses both the leakage concern and the practical need for a model that works when subscription data is unavailable at prediction time.

The risk assessment identified genuine deployment challenges — data pipeline fragility, the E-assist data gap, neighborhood equity concerns, and privacy governance requirements — that must be addressed systematically before this model moves from academic prototype to production system. Responsible deployment requires these issues to be resolved as a precondition, not as post-launch afterthoughts.

Reflectively, this project produced growth across multiple dimensions of data science practice. Technical skills in pipeline design, temporal splitting, and imbalanced classification were developed to a professional level. Process discipline in documentation, modular coding, and systematic evaluation was practiced and refined. Domain knowledge of urban mobility patterns, subscription service dynamics, and geographic concentration principles was acquired and will transfer to future transportation and GIS applications — connecting the analytical skills developed in this program to the spatial thinking developed across prior academic backgrounds.

The most durable lesson from this project is that analytical rigor and organizational relevance must be pursued simultaneously. A model that achieves perfect accuracy but cannot be explained to stakeholders, deployed responsibly, or adapted when conditions change does not fulfill the promise of data science. This project has been an exercise in building both the technical capability and the professional judgment to apply it in ways that create genuine, lasting value for the organizations and communities data scientists serve.

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